

Noah Elsayed

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Education

University of Calgary – Bachelor of Science in Mechanical Engineering, GPA: 3.7/4.0

Expected April 2027

Experience

Mechanical Engineering Intern, Canadian Natural Resources Ltd. – Calgary, AB

May 2025 – April 2026

- Quantified pipeline flow losses using simulations and hand calculations, then implemented drag-reducing agents to increase throughput, reducing operating costs by \$12M annually
- Automated performance analysis workflows using VBA, decreasing analyst workload by 10+ hours weekly
- Mitigated off-spec delivery issues by validating condensate composition through a 15-site testing program
- Diagnosed a failed in-line inspection (ILI) tool through root cause analysis and implemented a standardized pre-run checking system to prevent repeat failures

Production Engineering Summer Student, Whitecap Resources Ltd. – Calgary, AB

May 2024 – Aug 2024

- Achieved <5% pump sizing error across 30+ cases by developing and validating a model in Python using hydraulic performance calculations and field statistics
- Enabled accurate pump selection by deploying the validated sizing model as a user-facing application adopted by field engineers, used in 2-4 sizings monthly
- Identified root causes of pump failures through tear-downs and failure analysis, informing maintenance and pump selection strategy updates
- Saved 100+ hours annually by analyzing surface casing CO₂ leakage data and flagging anomalies with a geospatial Power BI dashboard

Engineering Summer Student, BRE Group – Calgary, AB

May 2023 – Aug 2023

- Developed fluid mechanics case studies incorporating thermodynamic and phase-behavior calculations for a carbon capture and storage (CCS) training program with 200+ participants
- Co-authored a technical article analyzing common reservoir simulation pitfalls affecting model accuracy

Projects

Formula SAE Drivetrain and Cooling System

Sept 2023 – Present

- Designed the team's first electric vehicle cooling system using Simulink by modeling heat generation and airflow requirements, establishing a validated framework for future development
- Enabled data-driven thermal design by developing an STM32-based data logging system in Altium to capture real-time drivetrain temperatures
- Lowered vehicle center of gravity by 7 mm by designing and finite element analysis-validating a motor controller mount to a 2.5× safety factor in SolidWorks, verifying results through manual calculations
- Manufactured drivetrain and cooling components using mill, lathe, and waterjet equipment, translating designs into track-ready hardware

Disease Prediction Tool

github.com/zzaid17/biohack-2025

- Built a medical pre-screening web application by developing and validating disease prediction models on 5 public health datasets
- Led a 4-person team to a 1st place finish and \$500 prize at Biohack 2025 hackathon

Skills

Engineering Design: SolidWorks, Inventor, Fusion, AutoCAD, Thermal Desktop, Simulink

Programming & Data Visualization: MATLAB, Tableau, Power BI, C++, Python, SQL, Git